

Customer Shopping Behavior Analysis Project

1. Project Introduction

This project analyzes customer shopping behavior using Python, SQL, and Power BI. The main goal is to understand customer purchasing trends, subscription behavior, discount usage, product ratings, and revenue generation.

2. Dataset Description

Rows	3900
Columns	18

Main Columns: Customer ID, Age, Gender, Item Purchased, Category, Purchase Amount (USD), Location, Size, Color, Season ...

3. Data Preprocessing and Cleaning using Python

Before analysis, the dataset was cleaned and preprocessed using Python and Pandas. The preprocessing steps include:

- Loading the CSV dataset using Pandas.
- Checking null values and duplicate records.
- Verifying data types.
- Cleaning column names.
- Handling missing values if present.
- Preparing data for SQL and Power BI analysis.

```
import pandas as pd

df = pd.read_csv("customer_shopping_behavior.csv")

# Check missing values
print(df.isnull().sum())

# Remove duplicates
df.drop_duplicates(inplace=True)

# Dataset information
print(df.info())
```

4. SQL Analysis

After preprocessing, the cleaned dataset was imported into PostgreSQL. SQL queries were used to generate business insights and customer behavior analysis.

- Revenue comparison between male and female customers.
- Top 5 products with highest review ratings.
- Discount impact on purchases.
- Customer segmentation into New, Returning, and Loyal customers.
- Top products in each category.
- Revenue contribution by age groups.
- Subscription vs non-subscription revenue analysis.

```
SELECT "Gender", SUM("Purchase Amount (USD)") AS Revenue
FROM customer
GROUP BY "Gender";
```

5. Power BI Dashboard

The SQL output was connected to Power BI to create interactive dashboards. The dashboard helps users understand customer behavior visually using charts, cards, filters, and KPIs. Dashboard Components: • Total Revenue Card • Average Purchase Amount • Average Review Rating • Revenue by Gender • Revenue by Age Group • Top Purchased Products • Subscription Status Analysis • Shipping Type Comparison

6. Key Insights

- Subscribed customers spend more on average.
- Discounts increase customer purchase frequency.
- Loyal customers generate higher revenue.
- Highly rated products receive repeat purchases.
- Certain age groups contribute more revenue.

7. Conclusion

This project successfully analyzed customer shopping behavior using Python, SQL, and Power BI. The analysis provides useful business insights that can help organizations improve marketing strategies, customer engagement, and sales performance.

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